

Figure 3.6 Regardless of which street grid is used, everyone agrees on the shape of a trail, in this case, a straight line.

In this sense, then, special relativity tells us that spacetime itself is the ultimate arbiter of accelerated motion. Spacetime provides the backdrop with respect to which something, like a spinning bucket, can be said to accelerate even in an otherwise empty universe. With this insight, the pendulum swung back again: from Leibniz the relationist to Newton the absolutist to Mach the relationist, and now back to Einstein, whose special relativity showed once again that the arena of reality—viewed as spacetime, not as space—*is* enough of a something to provide the ultimate benchmark for motion.[11]

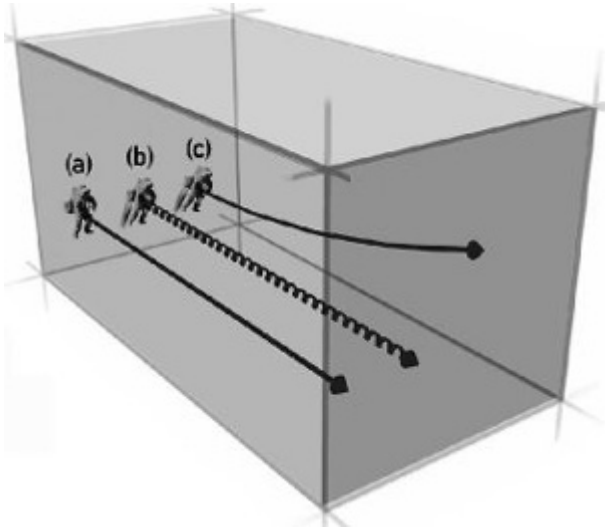


Figure 3.7 The paths through spacetime followed by three astronauts. Astronaut (a) does not accelerate and so follows a straight line through spacetime. Astronaut (b) flies repeatedly in a circle, and so follows a spiral through spacetime. Astronaut (c) accelerates into deep space, and so follows another curved trajectory in spacetime.